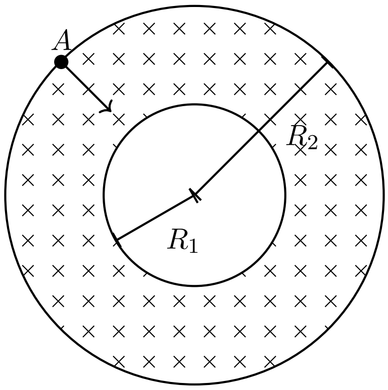
Electromagnetic in Disk

[5 pts]

Consider the annular region $R_1 \le r \le R_2$ in cylindrical coordinates (r, ϕ, z). A uniform magnetic field

$$\mathbf{B} = B\,\hat{\mathbf{k}},$$

directed upward, fills this region.



A particle of mass m and charge q enters the region at a point A on the outer circle $r = R_2$, with speed v directed radially inward toward the axis. The particle traces through a trajectory tangent to the inner circle $r = R_1$, subsequently leaves the region, and never enters it again.

Questions.

Q1. Find B in terms of m, q, v, R_1, and R_2.